Laser scanning and full-field birefringence microscopy by spectral encoding of the polarization

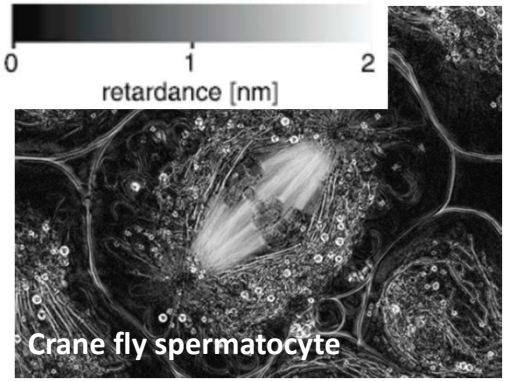
Hugo Laviec[a], Xavier Theillier[b], Matthieu Dubreuil[a], Sylvain Rivet[a] , Yann Le Grand[a]

a- Univ. Brest, OPTIMAG UR938

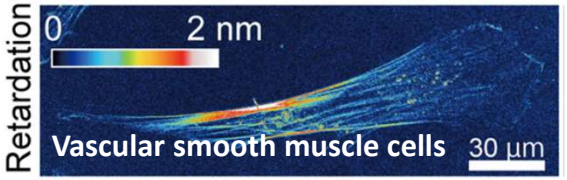
b- University of Limerick, Department of Physics and Bernal Institute

Mapping of optical retardation $\delta_{(x,y)} = \Delta ne$ (nm) and azimuth $\alpha_{(x,y)}$ (°)

Cells (microtubules, F-actin)



S. Metha et al, J. of Opt. (2013)



S. Sugita et al. Sci. Rep. 9 (2019)

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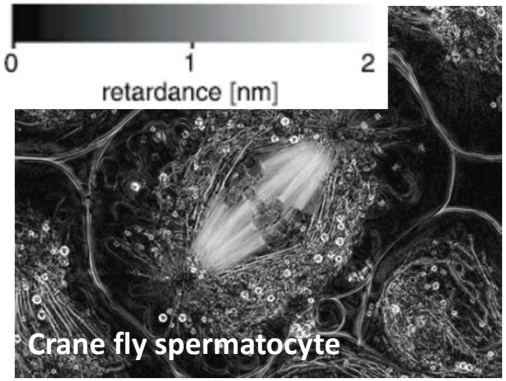

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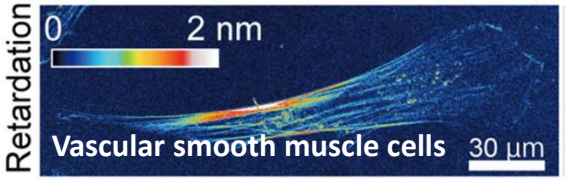
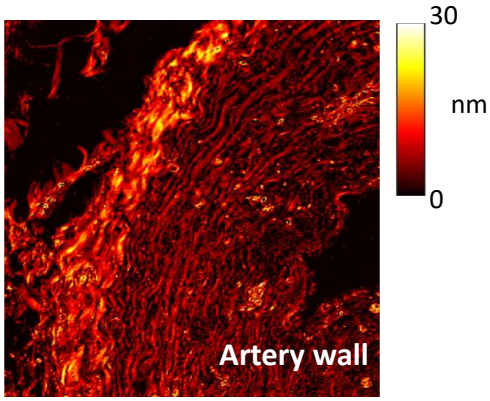


S. Sugita et al. Sci. Rep. 9 (2019)

ECM of tissues (collagen, elastin)

S. Aknoun et al. Opt. Comm. (2018)



M. Dubreuil et al. BOE (2019)

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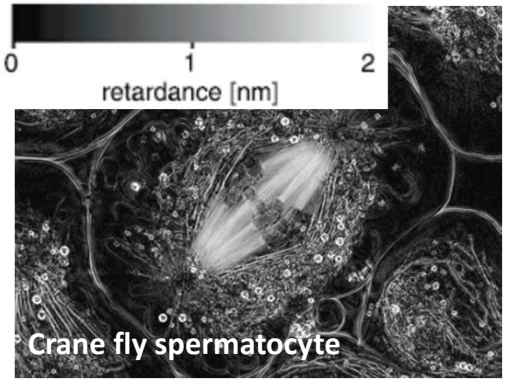

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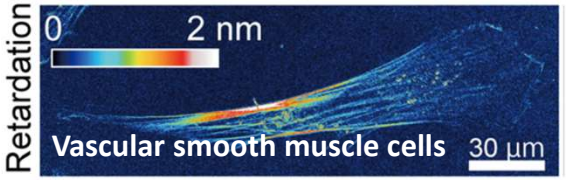

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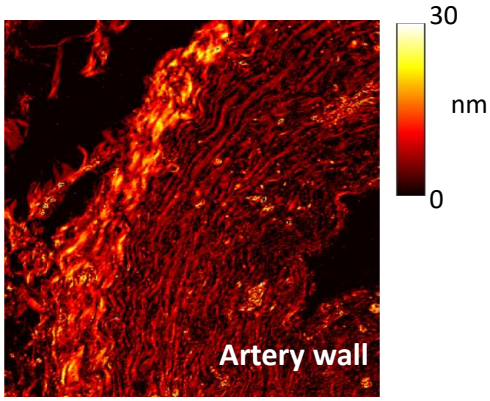

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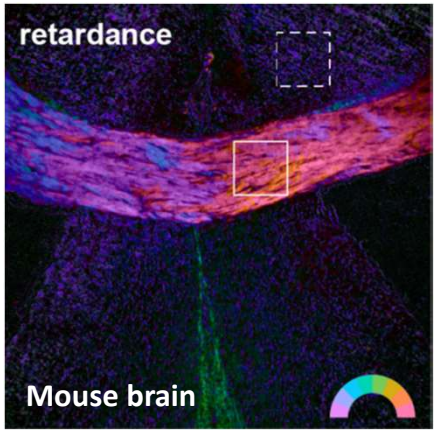
White matter (nervous fibers)



I. E. Ivanov et al. BOE (2022)

Full-field microscopy

- Most common microscopy method
- Active polarization states generation/analysis

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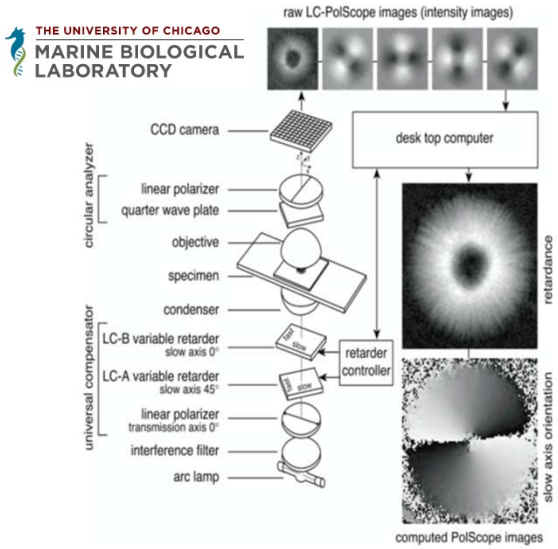

S. Mehta et al. J. of Opt. (2013)

Full-field microscopy

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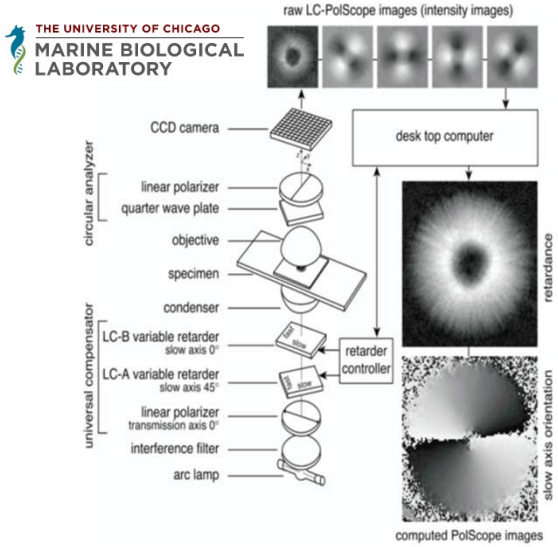

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Laser scanning microscopy

- Coupling with nonlinear imaging (SHG,TPEF)
- Fast passive polarization modulation (MHz)

Full-field microscopy

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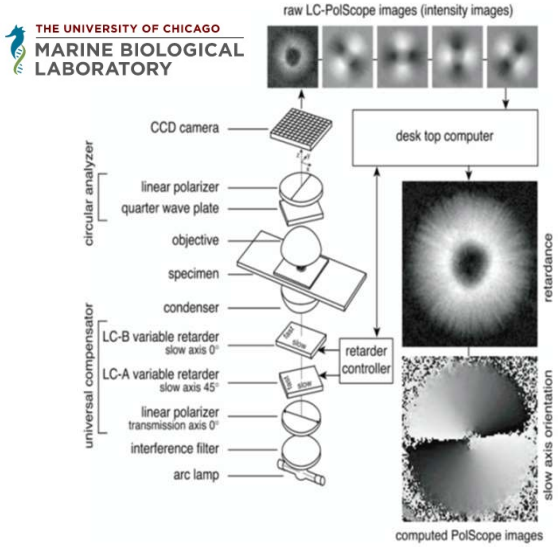

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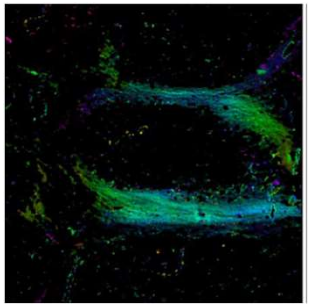
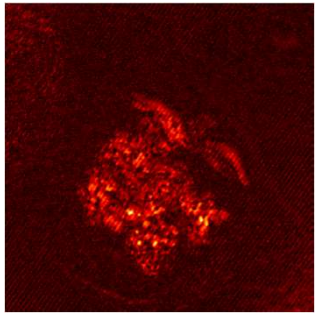
Spectral encoding



S. Rivet et al. Sci. Rep (2019)

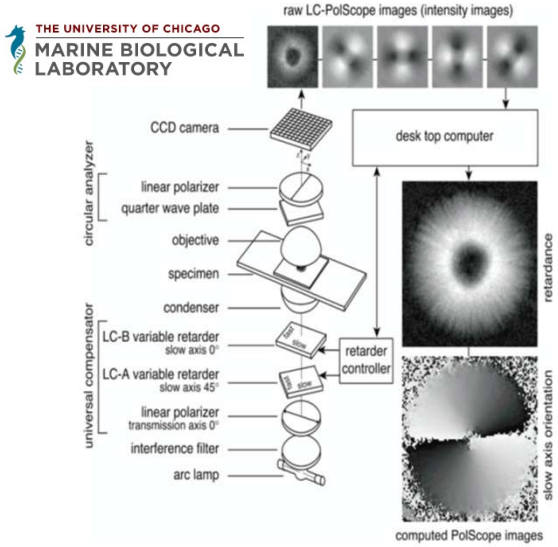
Orthogonality breaking



R. Desapogu et al. BOE (2021)

Full-field microscopy

- Most common microscopy method
- Active polarization states generation/analysis



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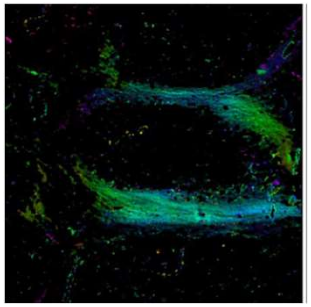

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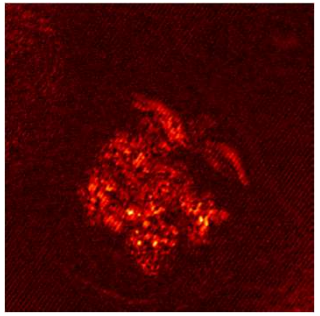

R. Desapogu et al. BOE (2021)

Challenges : acquisition time, sensitivity, exhaustivity, implementation in a measurement routine

Contents

I. Spectral encoding of the polarization

II. Gain in sensitivity: working close to extinction

III. High-sensitivity laser scanning birefringence microscopy by spectral encoding

IV. Transposition to full-field microscopy

I. **Spectral encoding of the polarization**

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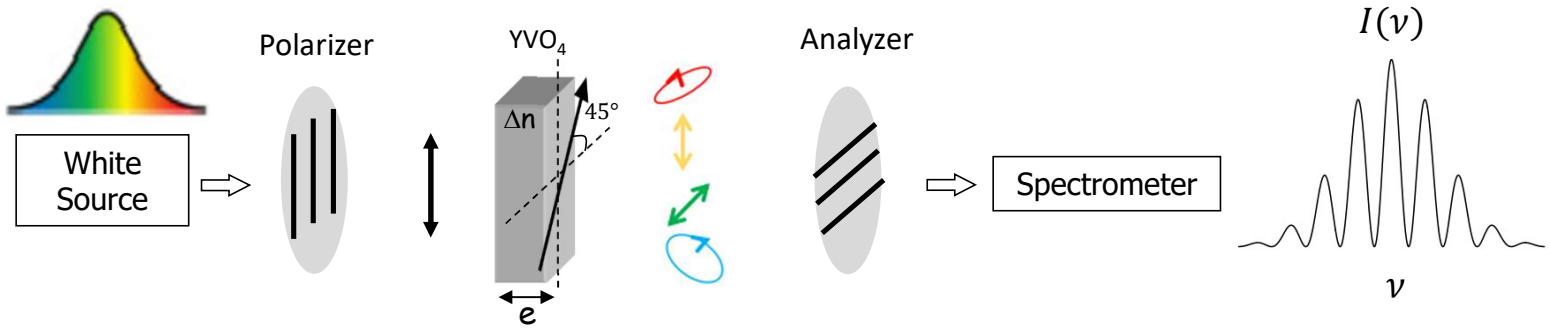
Spectral encoding of the polarization: two possibilities

M. Dubreuil *et al.*, **Opt. Express** (2007)



$$\varphi(\lambda) = \frac{2\pi\Delta n e}{\lambda}$$

Spectral encoding of the polarization: two possibilities

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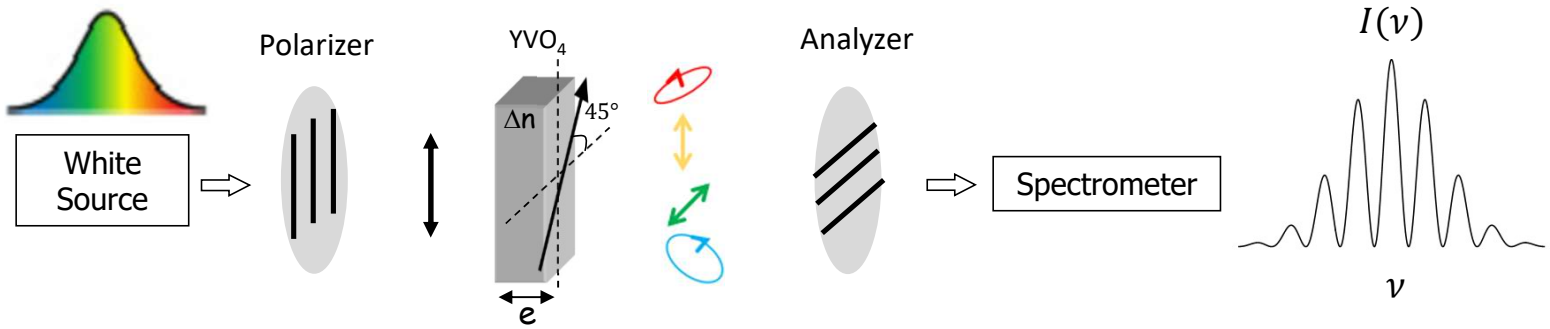

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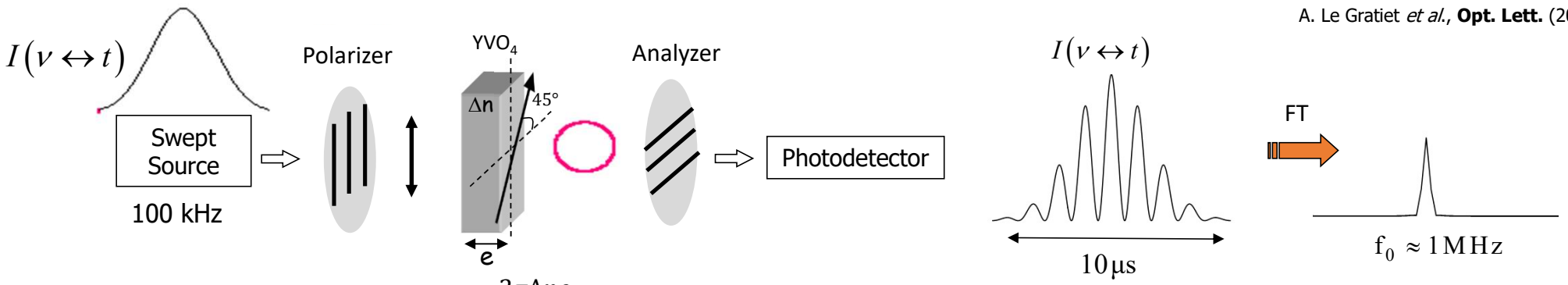
A. Le Gratiet *et al.*, **Opt. Lett.** (2015)



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High-frequency modulation

I. Spectral encoding of the polarization

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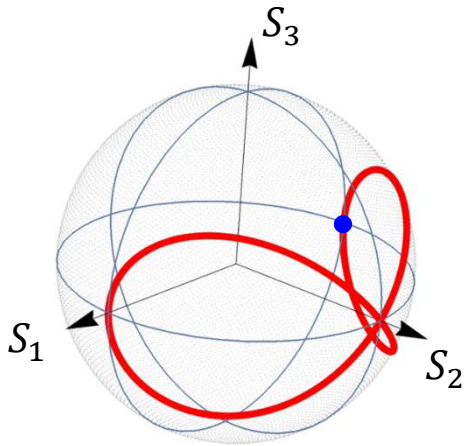
Increase sensitivity: working close to extinction



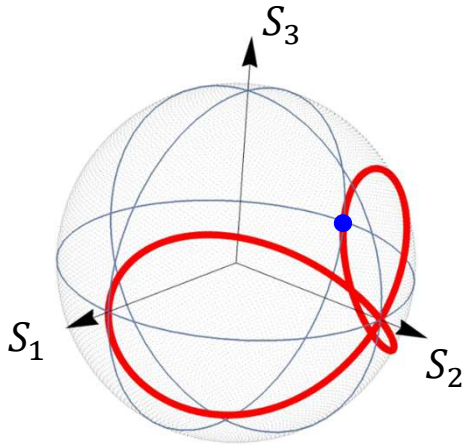
Polarization states incident on the specimen

Analyzer polarization state

Off-null (Mueller)

$Sensitivity \propto I^{sat}$ at best

Increase sensitivity: working close to extinction



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Full polarimetric characterization

Limited sensitivity

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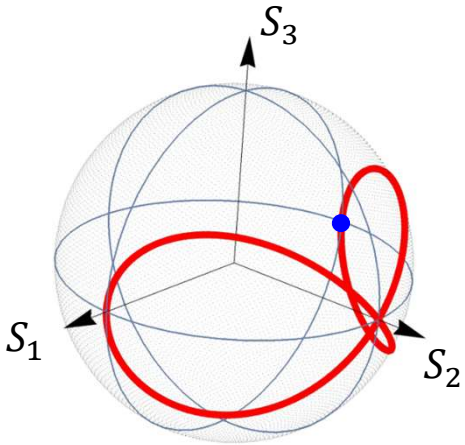

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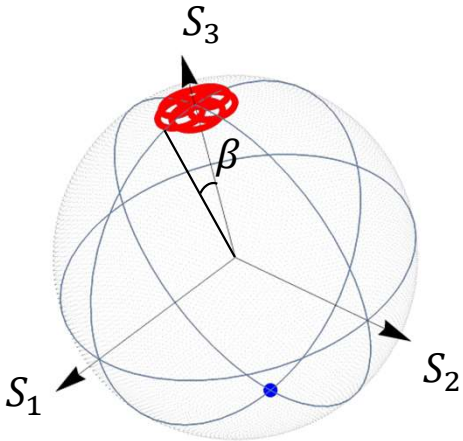

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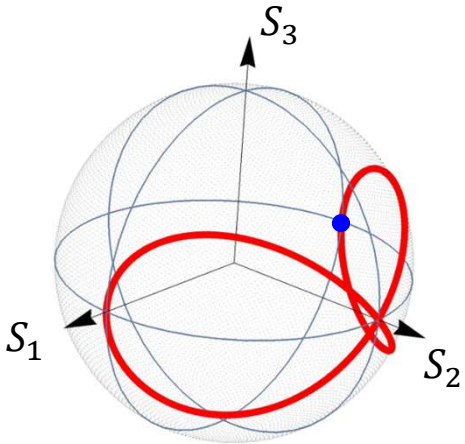

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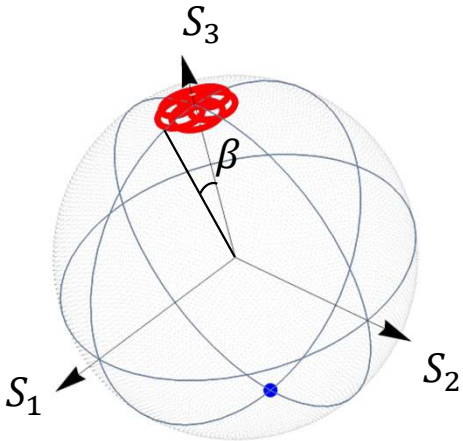

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$Sensitivity \propto I^{sat}$ at best

Full polarimetric characterization

Limited sensitivity

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$Sensitivity \propto \frac{I^{sat}}{\beta}$ gain in sensitivity for low β

High sensitivity

Partial polarimetric characterization (only birefringence or dichroism)

→ Hypothesis: thin linear birefringent specimens

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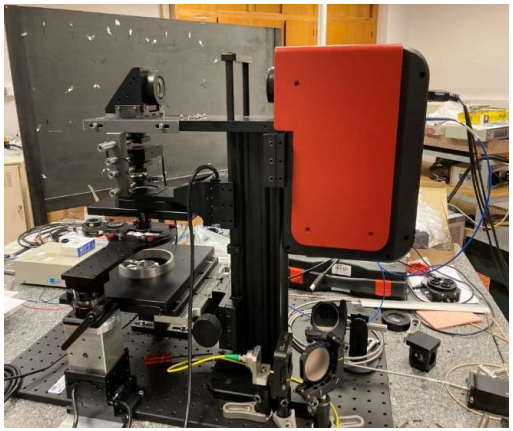
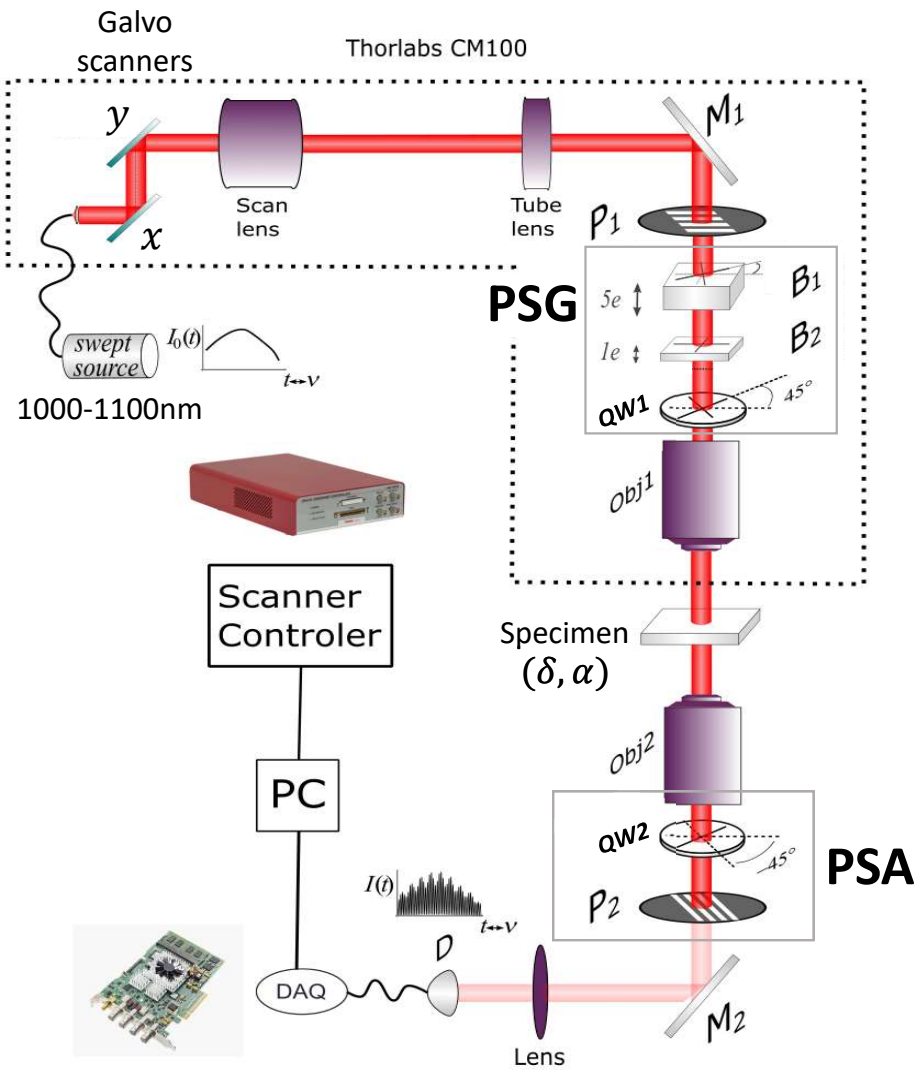
High-sensitivity laser scanning birefringence microscopy

X. Theillier *et al.*, **Opt. Express** (2022)
X. Theillier *et al.*, **Opt. Lett** (2024)





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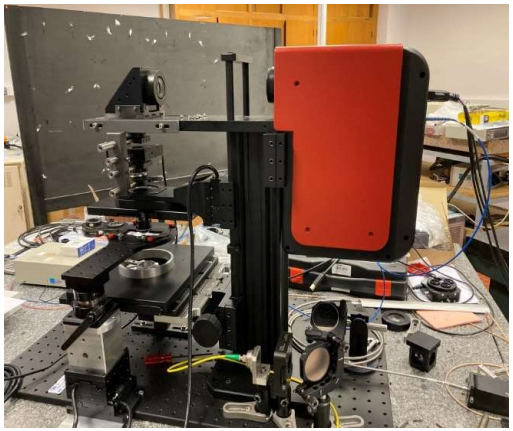

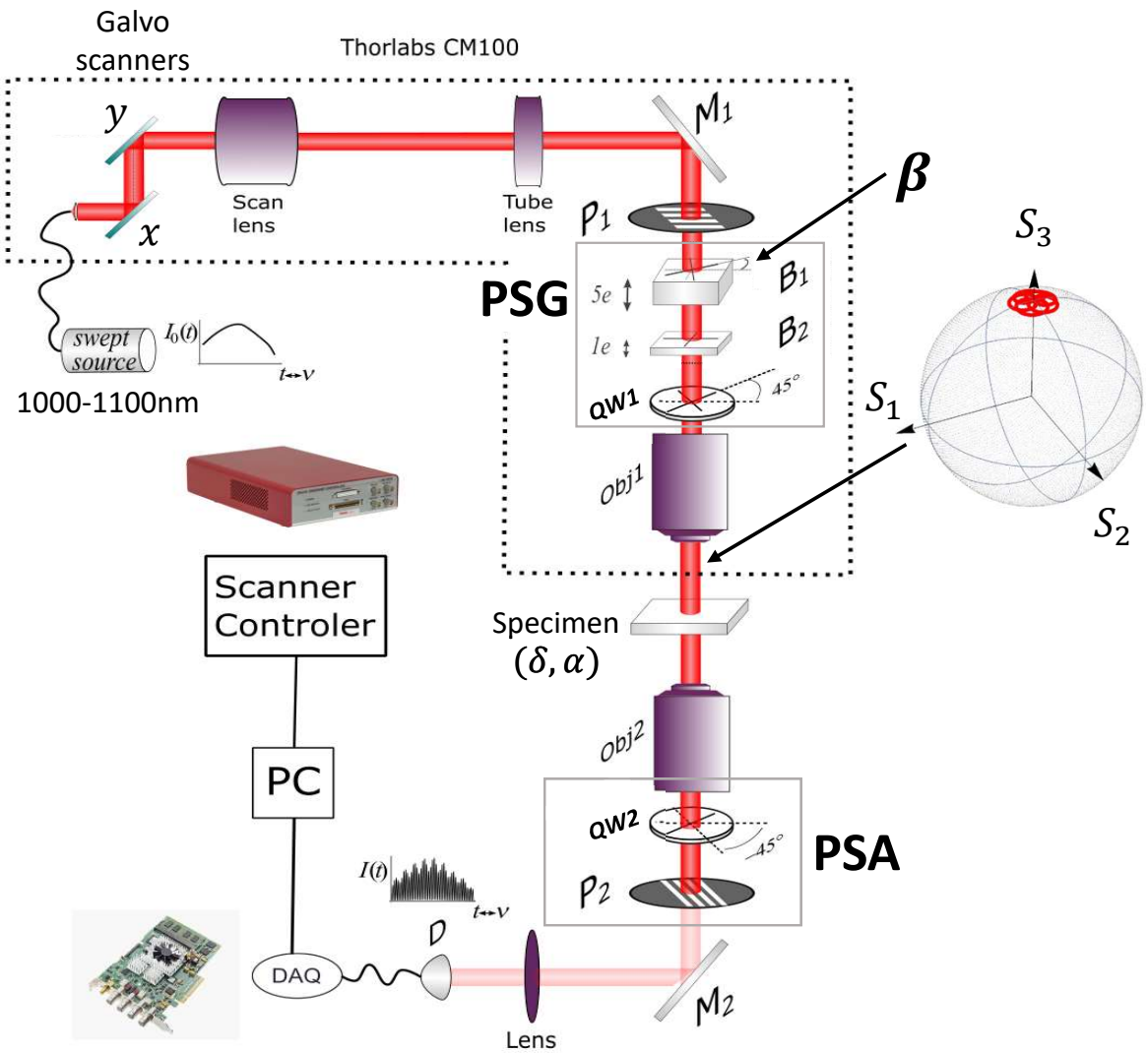

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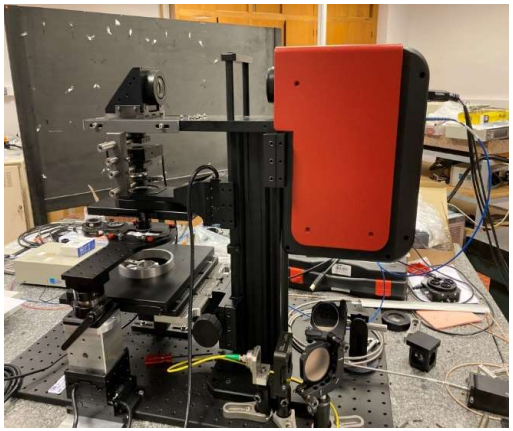
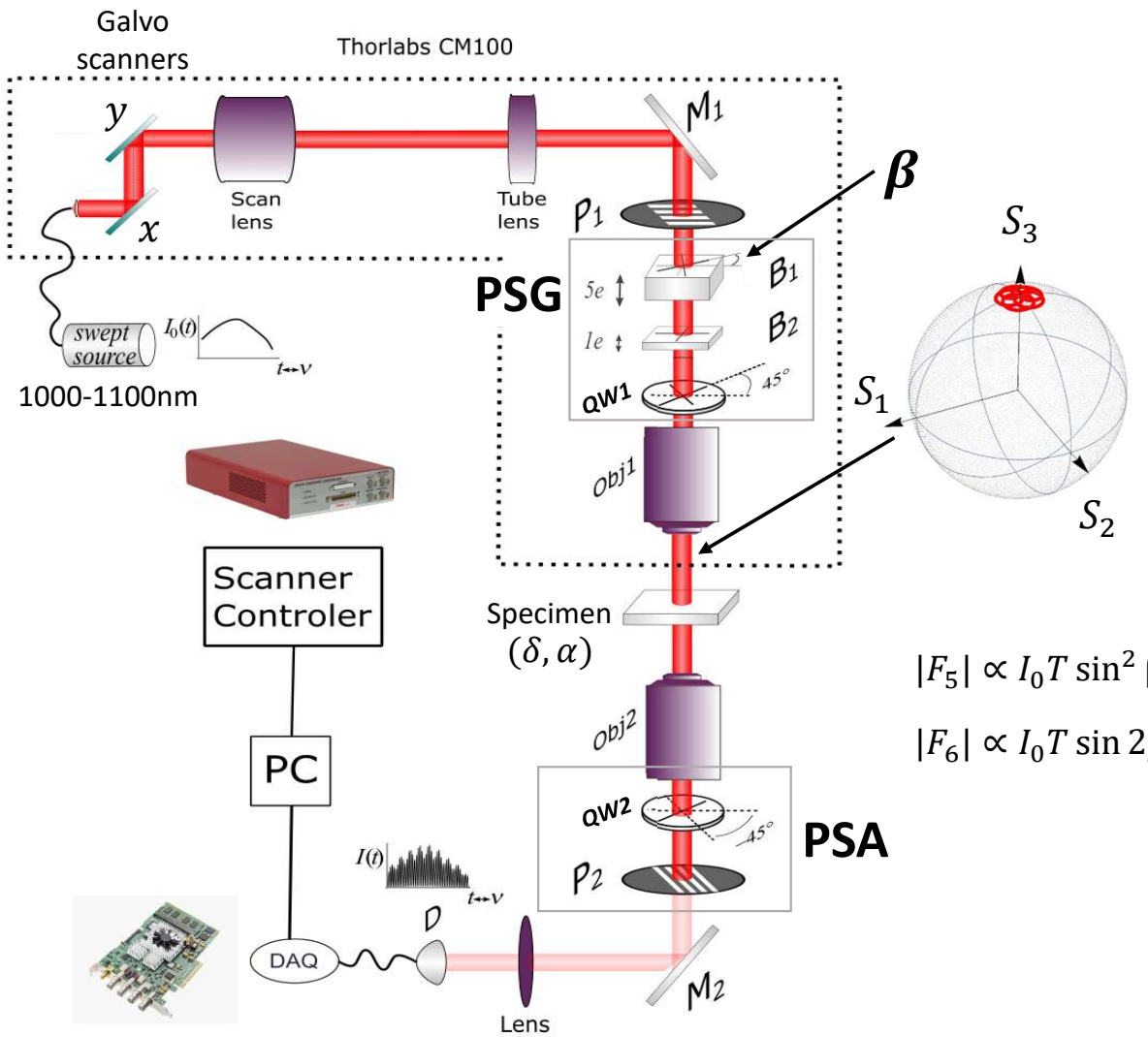
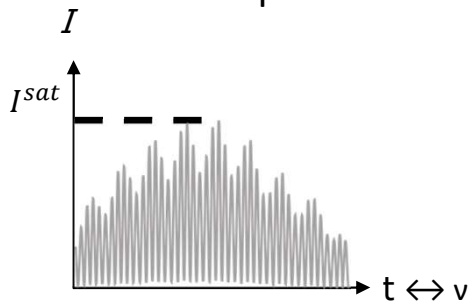




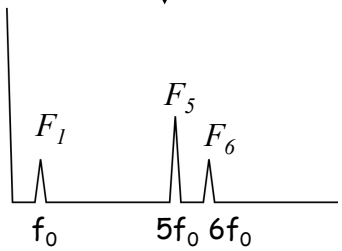
For one pixel



FT

$|F_5| \propto I_0 T \sin^2 \beta$

$|F_6| \propto I_0 T \sin 2\beta\, \delta$



$$\gamma = \frac{F_6}{F_5} \propto \frac{\delta}{2\tan\beta} e^{i2\alpha}$$

High-sensitivity laser scanning birefringence microscopy

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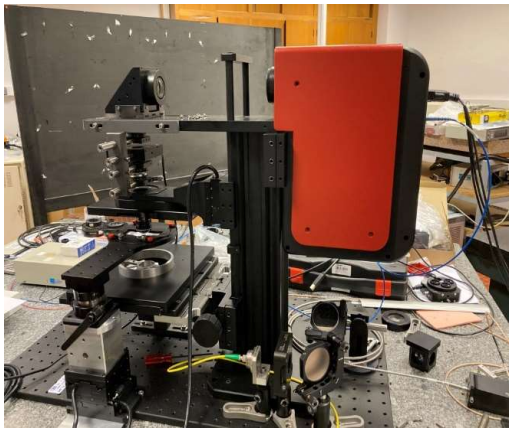
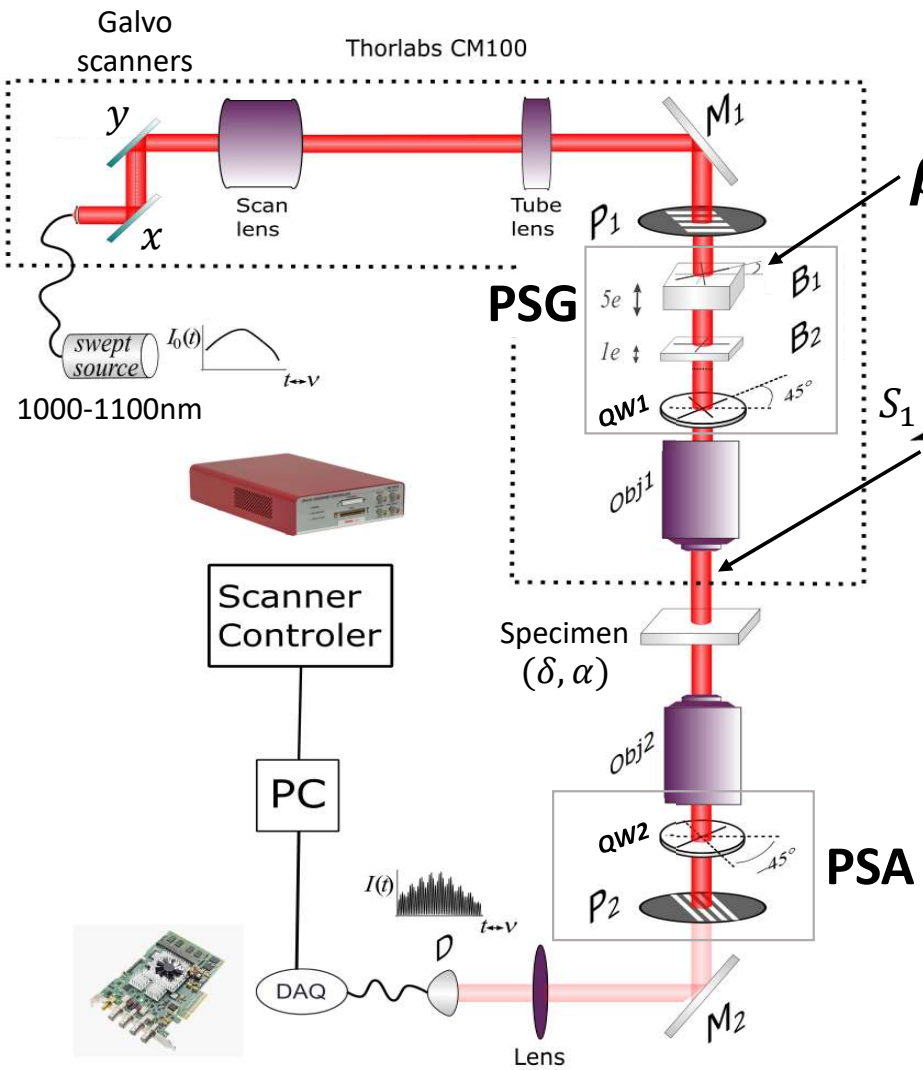
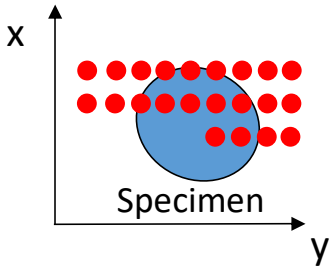
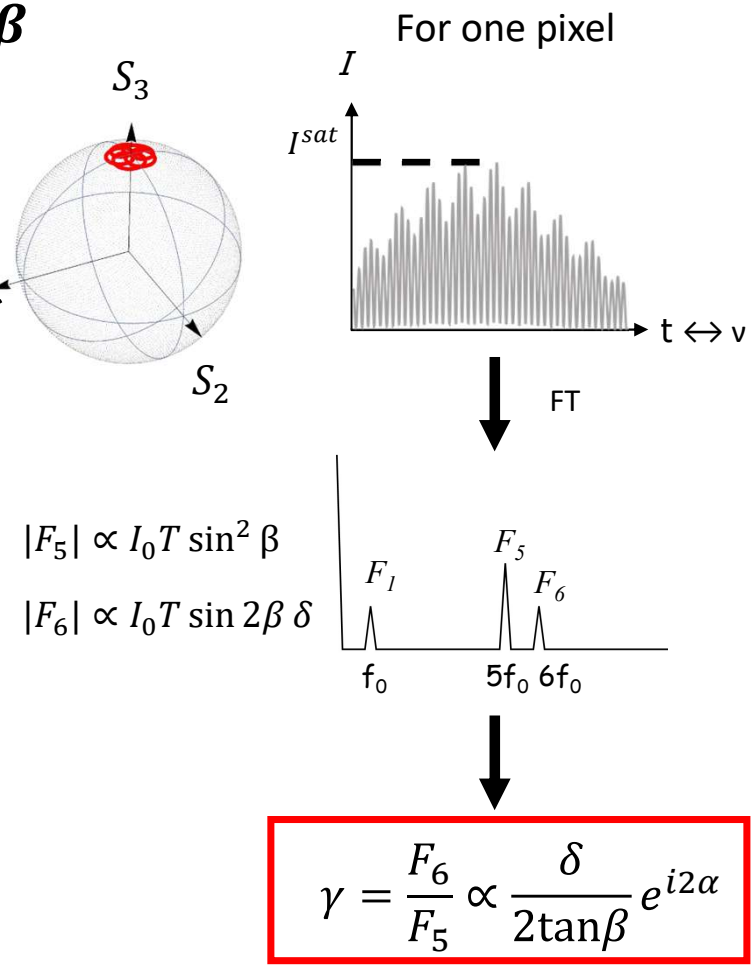






1 measure
= 10 µs
= 1 pixel



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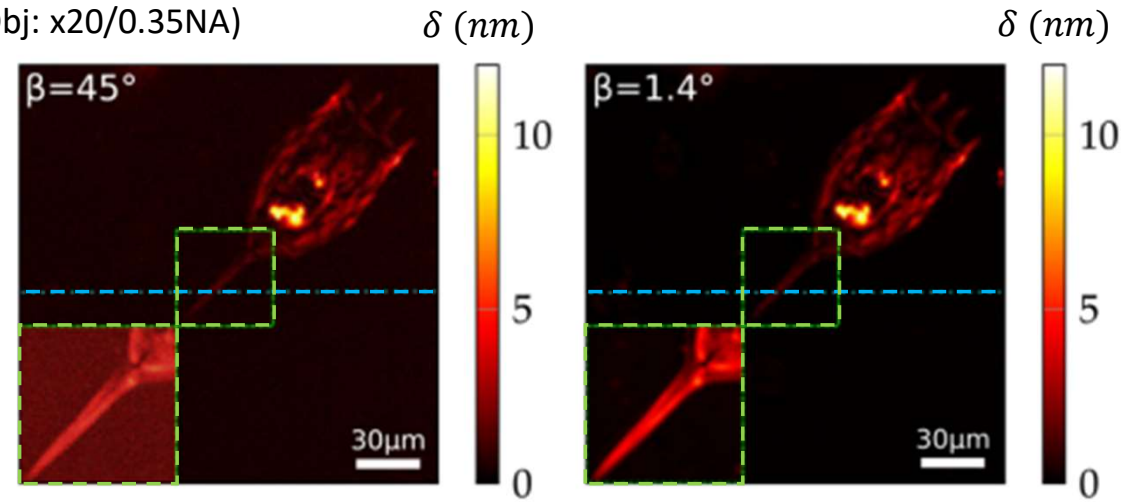
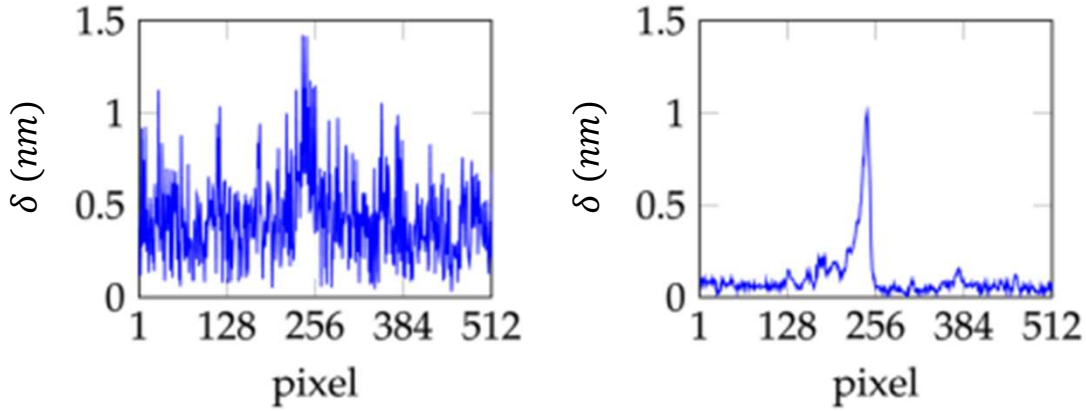
High-sensitivity laser scanning birefringence microscopy

Rotifer (Obj: x20/0.35NA)



(a) Off-null configuration **(b)** Null configuration

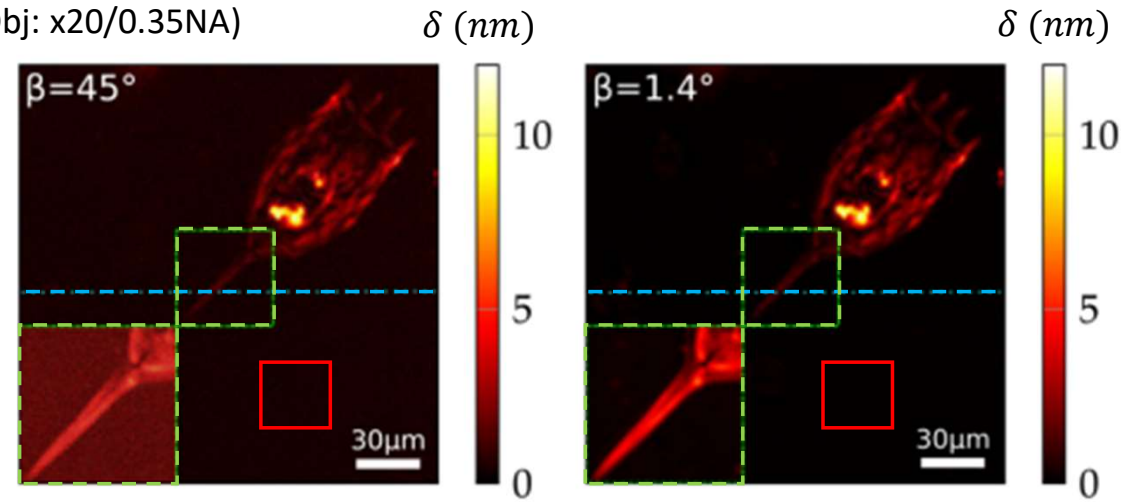


(c) Off-null configuration **(d)** Null configuration

X. Theillier *et al.*, **Opt. Lett** (2024)

High-sensitivity laser scanning birefringence microscopy

Rotifer (Obj: x20/0.35NA)



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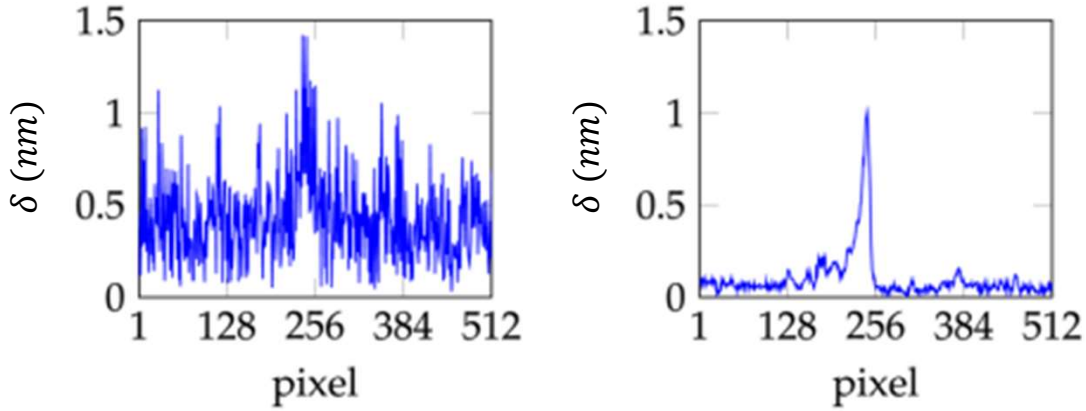


(c) Off-null configuration **(d)** Null configuration

X. Theillier *et al.*, **Opt. Lett** (2024)

Detection limit

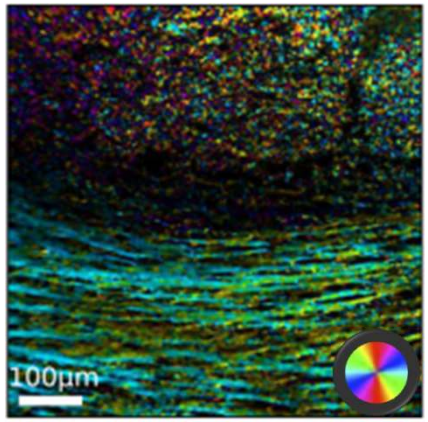
$$\beta = 45° \rightarrow \sigma_\delta = 240\ pm \leftrightarrow \frac{\lambda}{4000}$$

$$\beta = 1.4° \rightarrow \sigma_\delta = 8.5\ pm \leftrightarrow \frac{\lambda}{120\ 000}$$

σ_δ has decreased by a factor 30

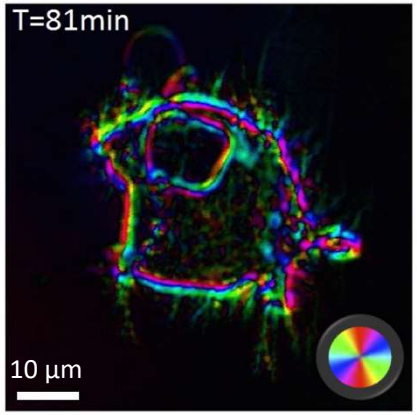
High-sensitivity laser scanning birefringence microscopy

X. Theillier *et al.*, **Opt. Lett** (2024)



Cancerous human colon tissue
(Obj: x10/0.25NA)

X. Theillier, **PhD Thesis** (2022)



Human gastric cancer cell HGT-1
(Obj: x60/1.20NA)

Lack of spatial resolution due to IR illumination → Full-field microscopy with illumination in the visible spectrum

I. Spectral encoding of the polarization

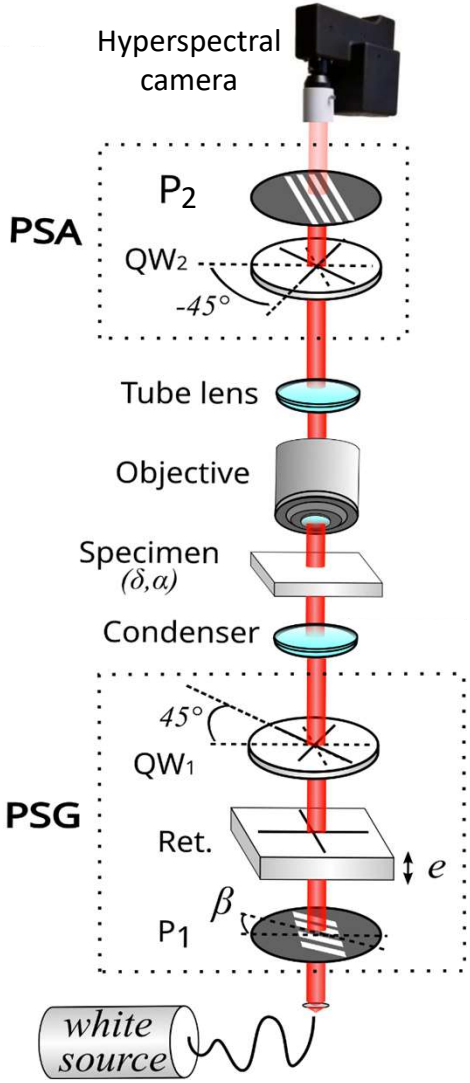
II. Gain in sensitivity: working close to extinction

III. High-sensitivity laser scanning birefringence microscopy by spectral encoding

IV. Transposition to full-field microscopy

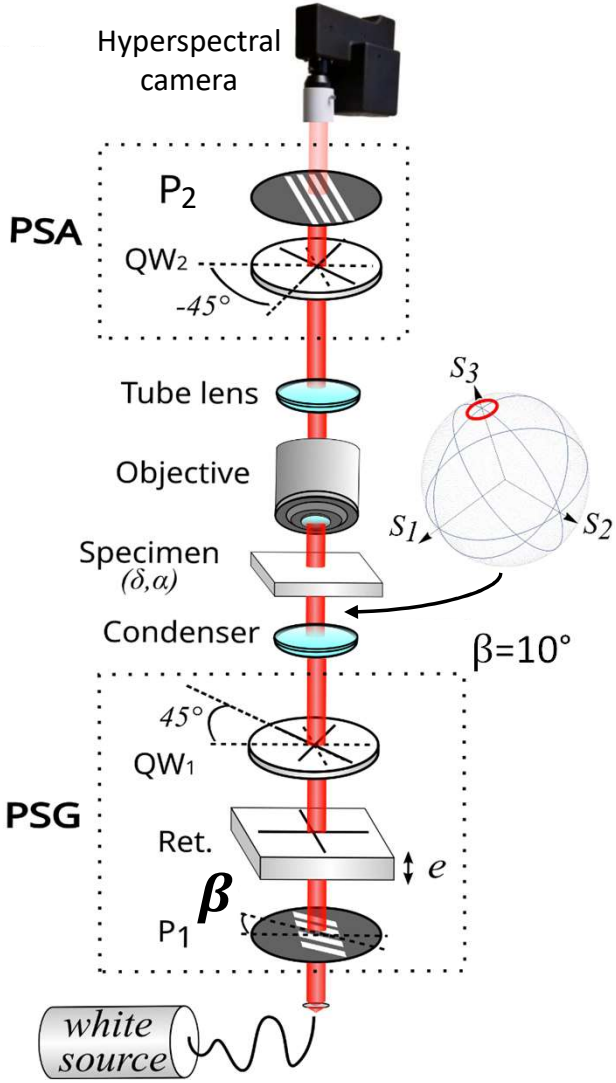
Birefringence imaging close to extinction by spectral coding with hyperspectral camera

H. Laviec *et al.*, *in preparation* for **Opt. Lett.** (2024)



Birefringence imaging close to extinction by spectral coding with hyperspectral camera

H. Laviec *et al.*, *in preparation* for **Opt. Lett.** (2024)



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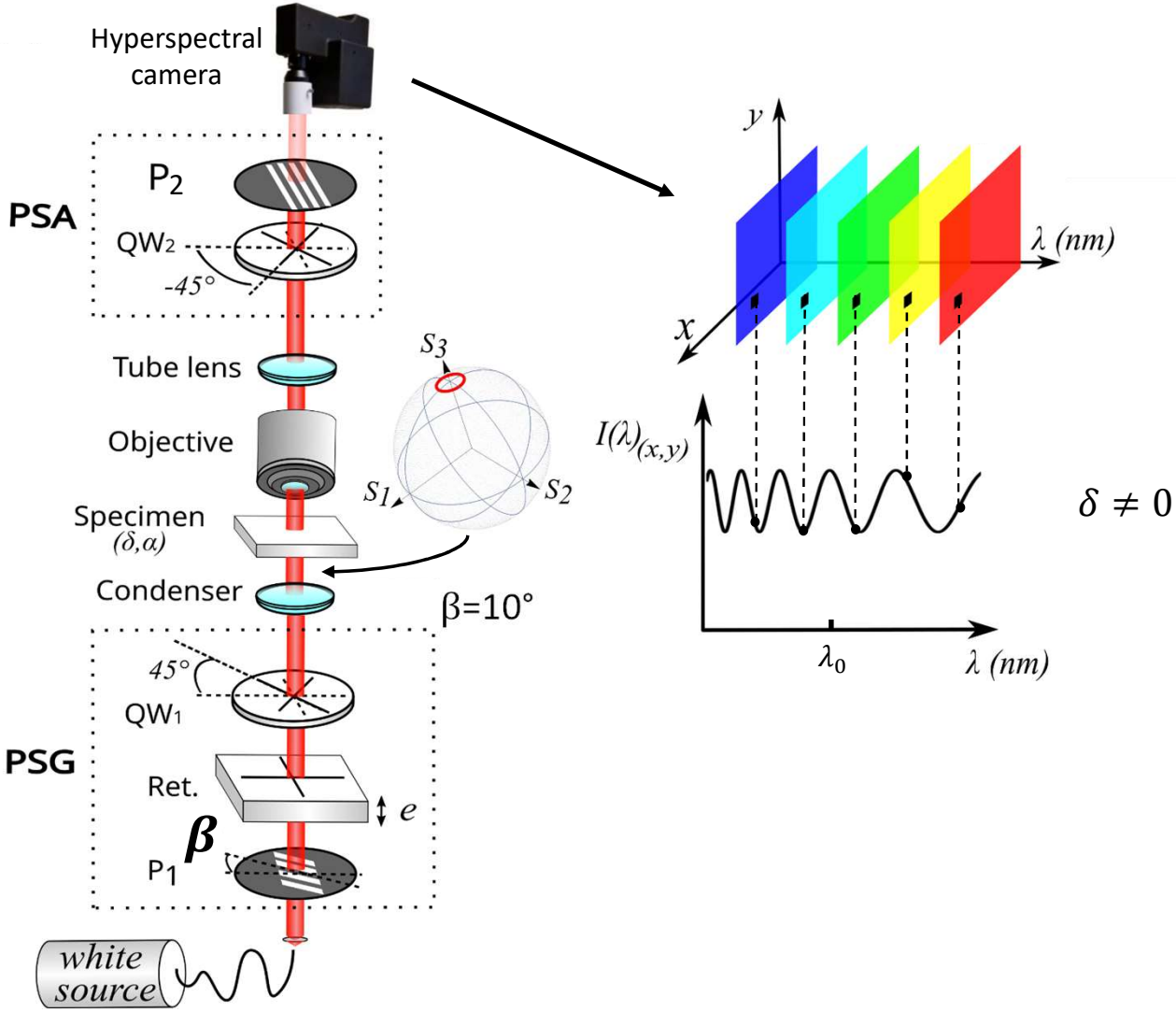

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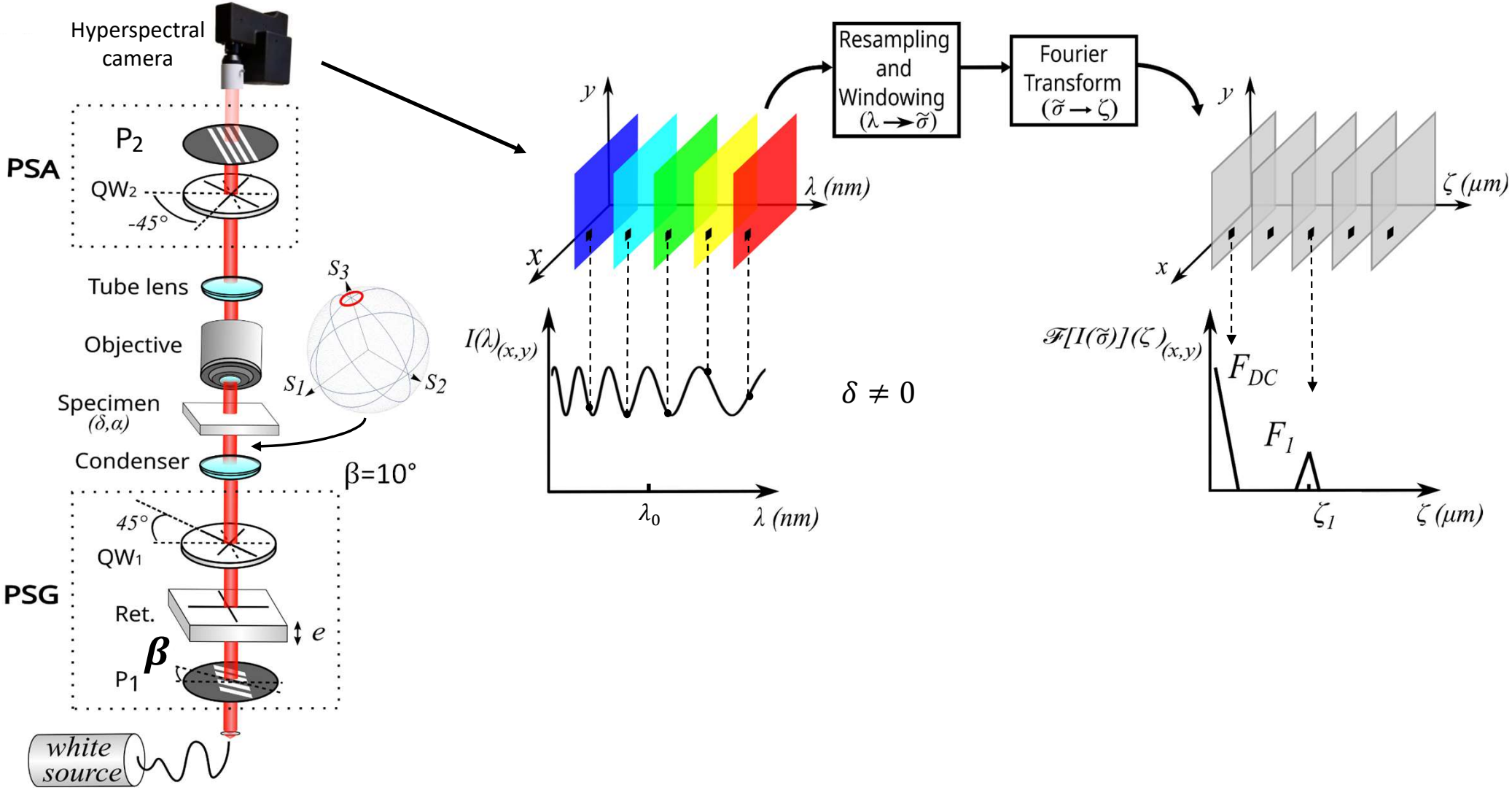
H. Laviec *et al.*, *in preparation* for **Opt. Lett.** (2024)



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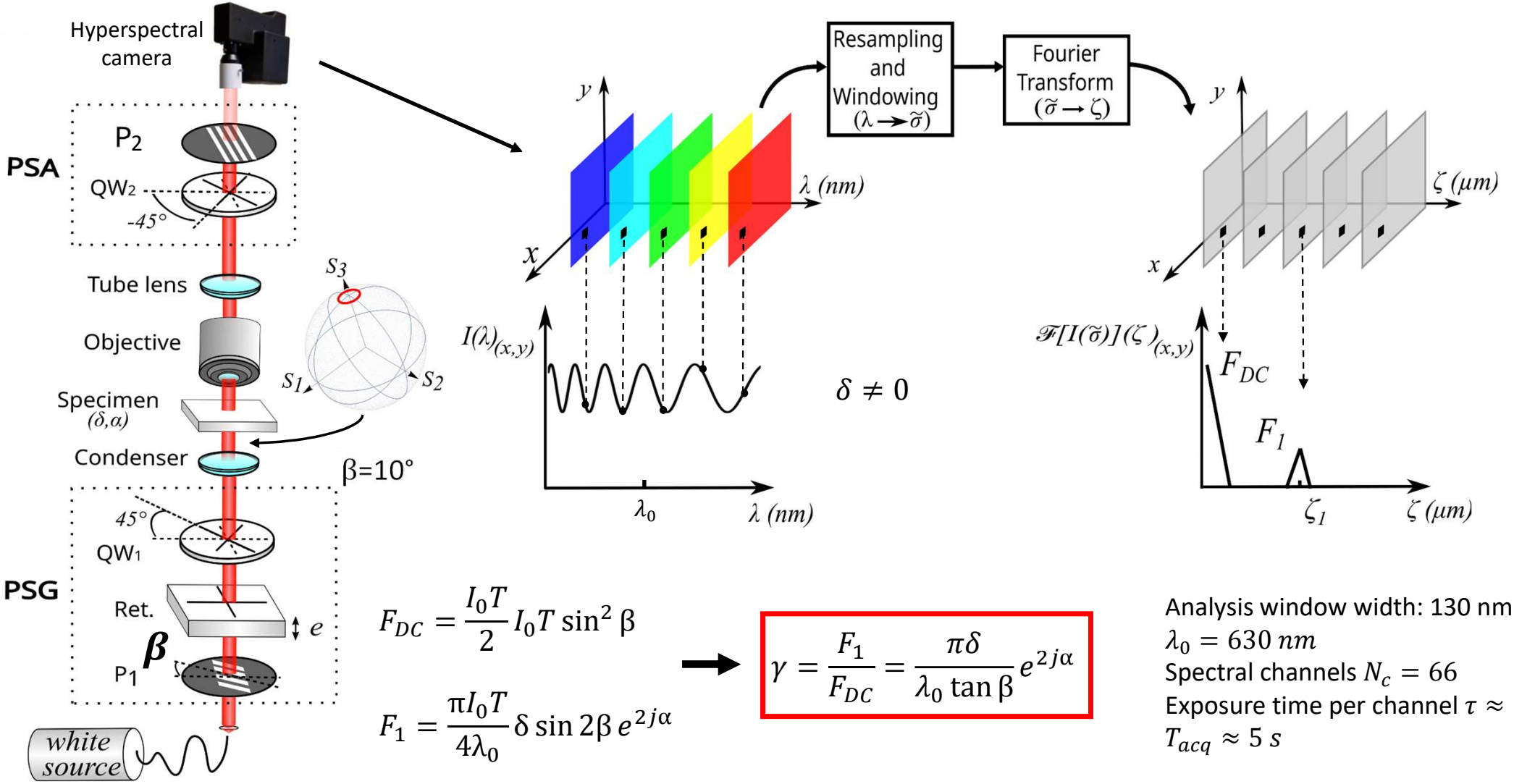
H. Laviec *et al.*, *in preparation* for **Opt. Lett.** (2024)



$$F_{DC} = \frac{I_0 T}{2} I_0 T \sin^2 \beta$$

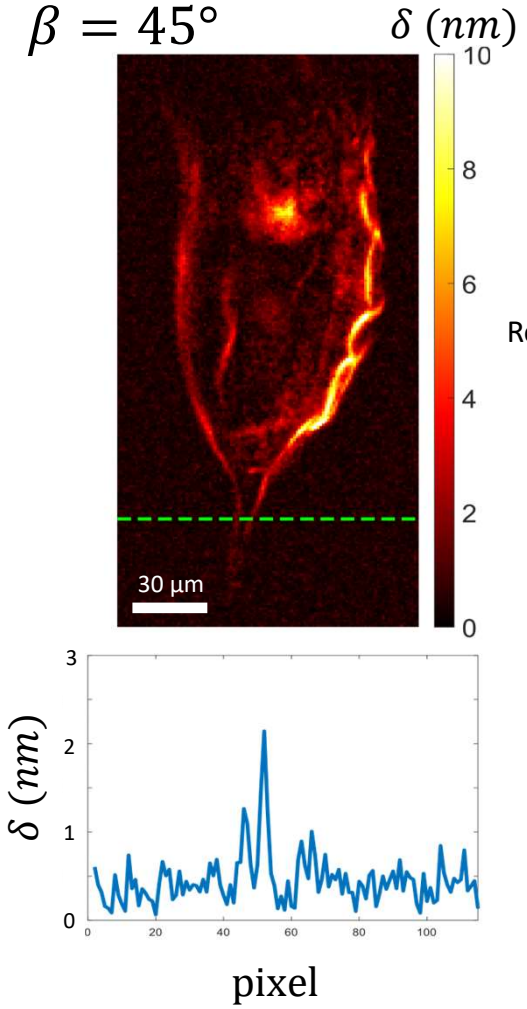
$$F_1 = \frac{\pi I_0 T}{4\lambda_0} \delta \sin 2\beta \, e^{2j\alpha}$$

$$\boxed{\gamma = \frac{F_1}{F_{DC}} = \frac{\pi\delta}{\lambda_0 \tan \beta} e^{2j\alpha}}$$

Analysis window width: 130 nm
$\lambda_0 = 630\ nm$
Spectral channels $N_c = 66$
Exposure time per channel $\tau \approx 100\ ms$
$T_{acq} \approx 5\ s$

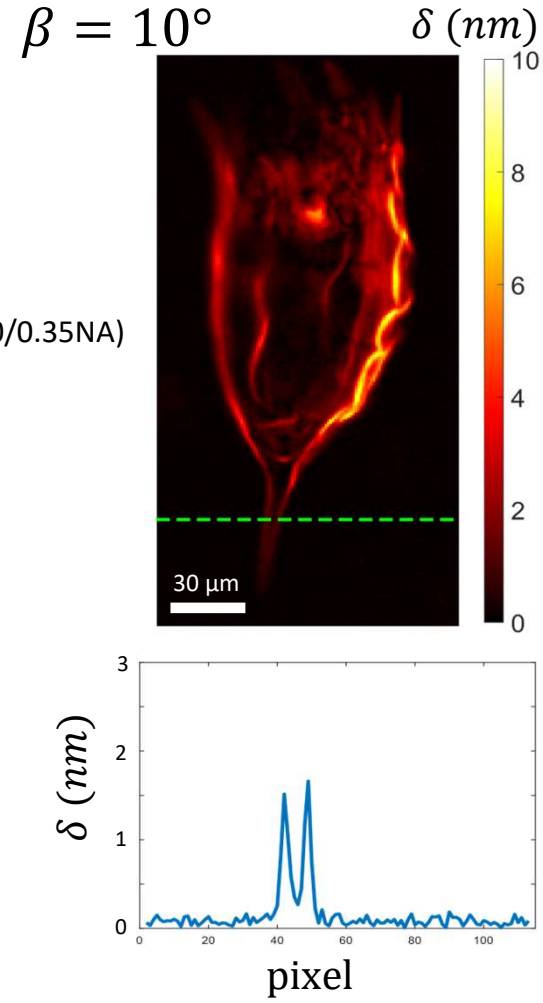
Birefringence imaging close to extinction by spectral coding with hyperspectral camera

$\beta = 45°$



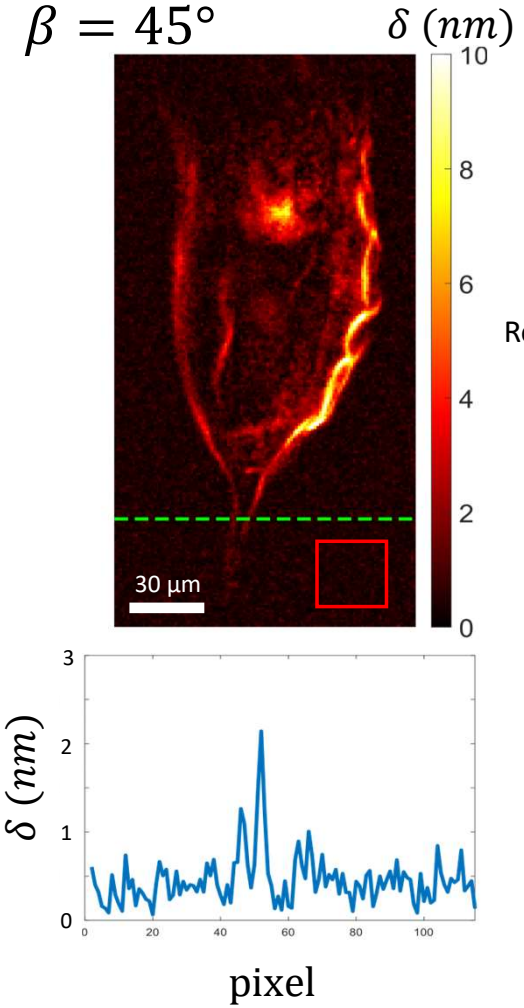
Rotifer (Obj: x20/0.35NA)

$\beta = 10°$



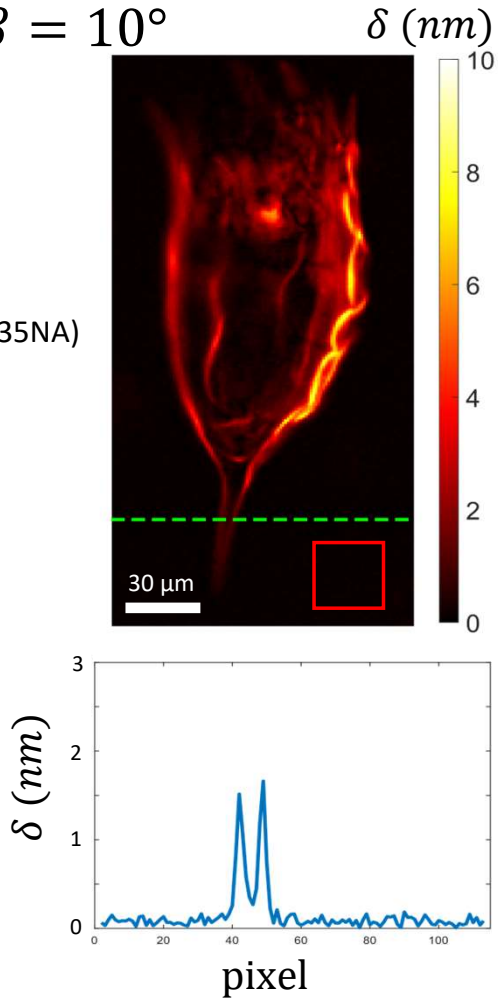
Birefringence imaging close to extinction by spectral coding with hyperspectral camera

$\beta = 45°$



Rotifer (Obj: x20/0.35NA)

$\beta = 10°$

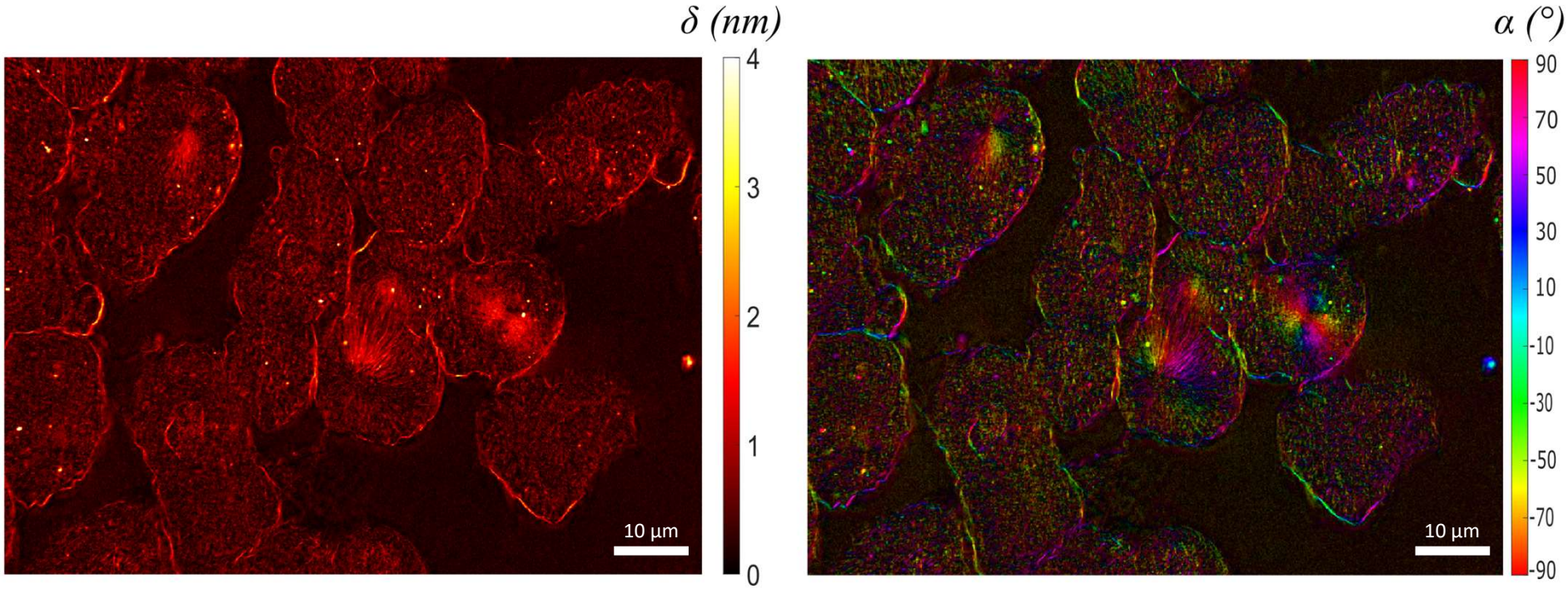


Detection limit

$$\beta = 45° \rightarrow \sigma_\delta = 220\ pm \leftrightarrow \frac{\lambda_0}{3000}$$

$$\beta = 10° \rightarrow \sigma_\delta = 45\ pm \leftrightarrow \frac{\lambda_0}{14\ 000}$$

Birefringence imaging close to extinction by spectral coding with hyperspectral camera



Whitefish cells (Obj: x60/1.4NA water immersion)

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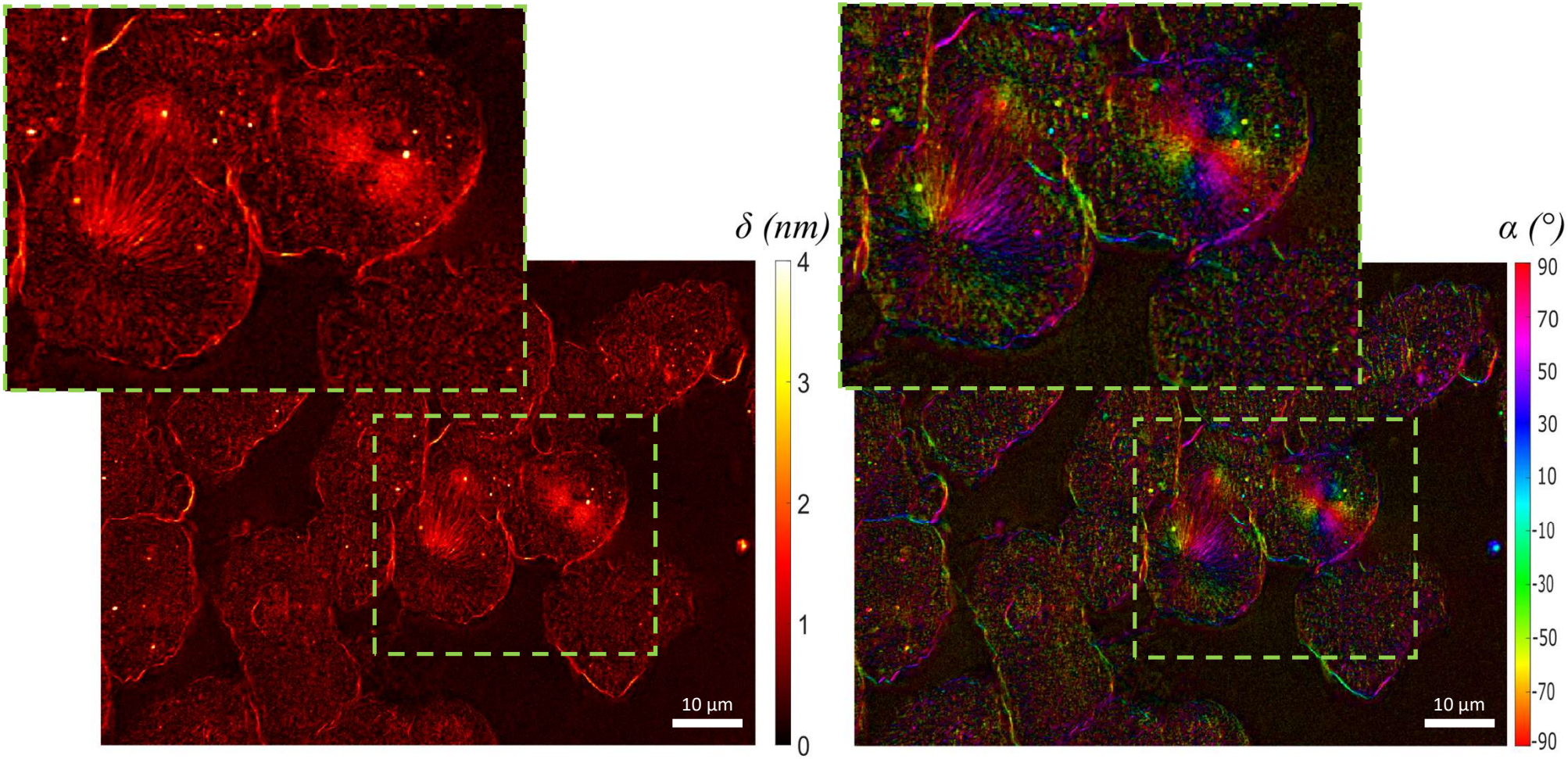
Birefringence imaging close to extinction by spectral coding with hyperspectral camera



Whitefish cells (Obj: x60/1.4NA water immersion)

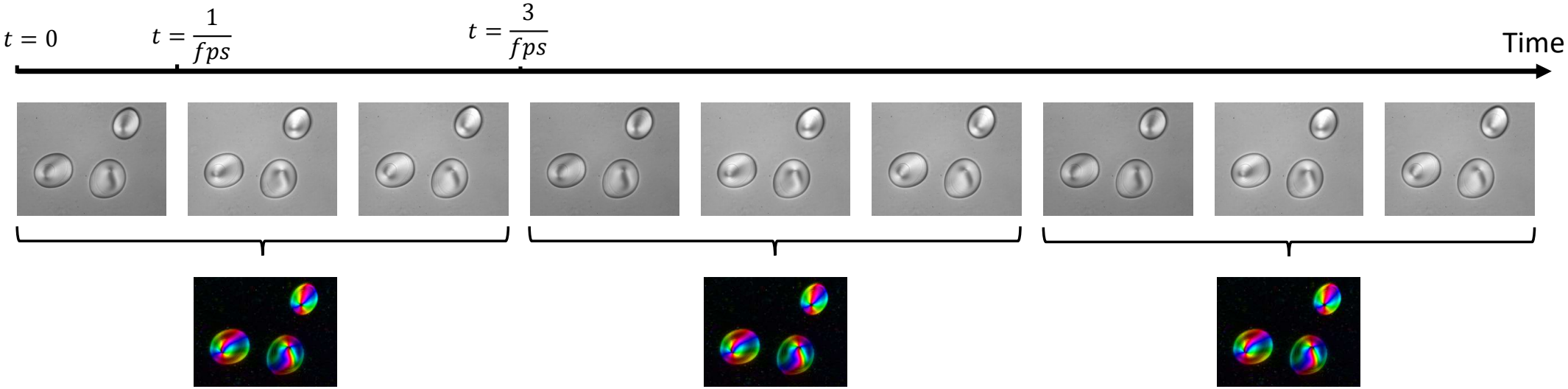
H. Laviec *et al.*, *in preparation* for **Opt. Lett.** (2024)

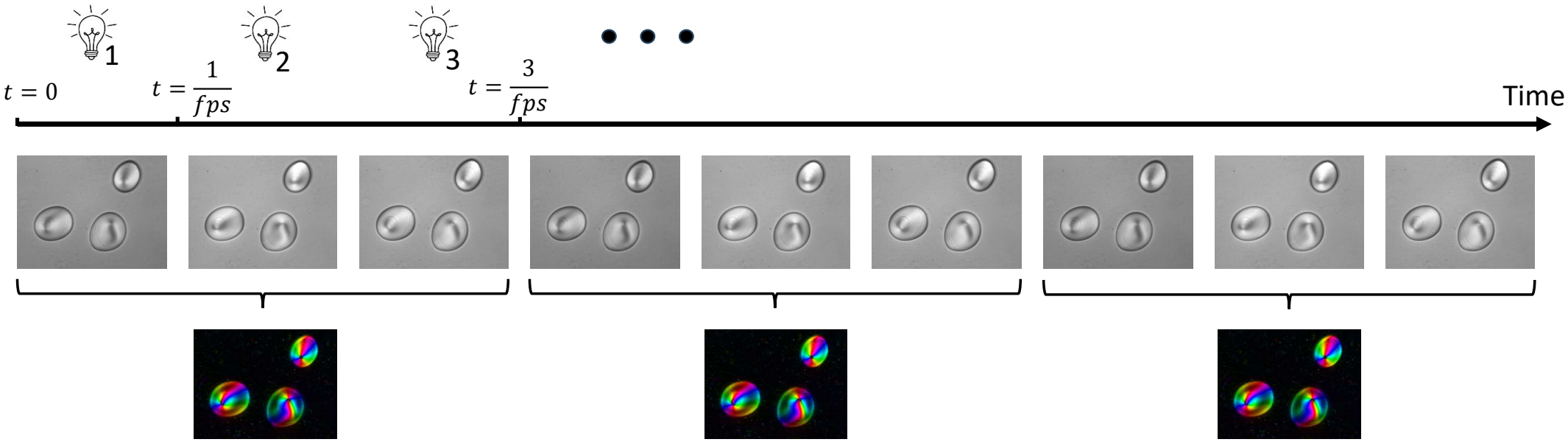
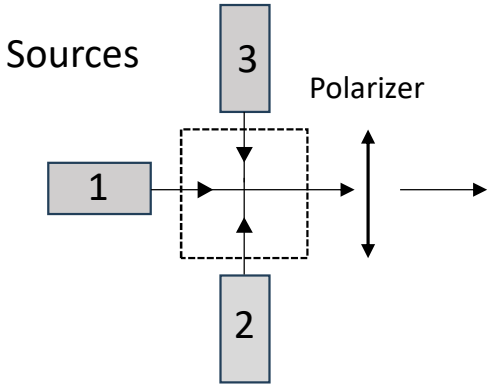
Conclusion

- Laser scanning microscopy: limited spatial resolution (IR source) → illumination in the visible spectrum
- Full-field microscopy: slow acquisition time (hyperspectral camera)

Perspectives

New demodulation approach using 3 images and high acquisition speed camera to realize full-field birefringence imaging in the visible at high speed (>30 fps) and high sensitivity (<0.01 nm)



Thank you for your attention

Conclusion and perspectives